



Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - I3

Time : 3 Hrs. 20 Mins.

Answers

1. (2)	41. (4)	81. (1)	121. (3)	161. (3)
2. (1)	42. (4)	82. (1)	122. (4)	162. (3)
3. (1)	43. (1)	83. (2)	123. (1)	163. (2)
4. (4)	44. (1)	84. (2)	124. (1)	164. (3)
5. (1)	45. (1)	85. (1)	125. (1)	165. (4)
6. (3)	46. (2)	86. (1)	126. (1)	166. (3)
7. (1)	47. (4)	87. (4)	127. (3)	167. (3)
8. (4)	48. (1)	88. (3)	128. (3)	168. (2)
9. (2)	49. (1)	89. (3)	129. (1)	169. (3)
10. (1)	50. (4)	90. (3)	130. (1)	170. (4)
11. (2)	51. (1)	91. (3)	131. (1)	171. (4)
12. (2)	52. (2)	92. (1)	132. (3)	172. (4)
13. (3)	53. (4)	93. (3)	133. (1)	173. (1)
14. (2)	54. (1)	94. (4)	134. (3)	174. (3)
15. (4)	55. (2)	95. (1)	135. (4)	175. (4)
16. (4)	56. (1)	96. (2)	136. (4)	176. (2)
17. (2)	57. (2)	97. (4)	137. (3)	177. (4)
18. (4)	58. (1)	98. (3)	138. (1)	178. (1)
19. (4)	59. (3)	99. (1)	139. (4)	179. (2)
20. (2)	60. (2)	100. (3)	140. (4)	180. (1)
21. (2)	61. (3)	101. (1)	141. (3)	181. (2)
22. (2)	62. (3)	102. (2)	142. (4)	182. (1)
23. (4)	63. (1)	103. (4)	143. (3)	183. (1)
24. (1)	64. (4)	104. (3)	144. (1)	184. (2)
25. (4)	65. (2)	105. (1)	145. (3)	185. (1)
26. (1)	66. (2)	106. (1)	146. (4)	186. (2)
27. (1)	67. (2)	107. (4)	147. (1)	187. (2)
28. (3)	68. (2)	108. (1)	148. (3)	188. (1)
29. (4)	69. (2)	109. (4)	149. (1)	189. (2)
30. (1)	70. (4)	110. (4)	150. (4)	190. (4)
31. (1)	71. (4)	111. (2)	151. (3)	191. (4)
32. (1)	72. (1)	112. (3)	152. (2)	192. (3)
33. (1)	73. (4)	113. (4)	153. (4)	193. (4)
34. (4)	74. (3)	114. (2)	154. (1)	194. (2)
35. (1)	75. (3)	115. (4)	155. (4)	195. (2)
36. (4)	76. (2)	116. (4)	156. (1)	196. (2)
37. (4)	77. (3)	117. (4)	157. (1)	197. (3)
38. (2)	78. (3)	118. (1)	158. (4)	198. (2)
39. (3)	79. (4)	119. (4)	159. (3)	199. (1)
40. (1)	80. (1)	120. (4)	160. (2)	200. (2)



Medical | IIT-JEE | Foundations

Corporate Office: Aakash Tower, 8, Pusa Road, New Delhi-110005, Ph.011-47623456

FINAL TEST SERIES for NEET-2024

MM : 720

Test - I3

Time : 3 Hrs. 20 Mins.

Answers and Solutions**PHYSICS****SECTION - A**

1. Answer (2)

Waves	Wavelength
AM radio waves	10^2 m
Microwaves	10^{-2} m
Infrared radiations	10^{-4} m
X-rays	10^{-10} m

(a) - (ii)

(b) - (iii)

(c) - (iv)

(d) - (i)

2. Answer (1)

$$F_{\text{up}} = 2000g + 3000$$

$$= 23000 \text{ N}$$

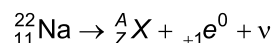
$$\text{Minimum power } P_{\text{min}} = \vec{F} \cdot \vec{v}$$

$$P_{\text{min}} = Fv = 23000 \times \frac{3}{2}$$

$$= 34500 \text{ W}$$

3. Answer (1)

The nuclear reaction is given as



From conservation of atomic number

$$11 = Z + 1 \Rightarrow Z = 10 \Rightarrow \text{Ne}$$

From conservation of mass number

$$22 = A + 0 \Rightarrow A = 22$$

$$\therefore {}_Z^AX = {}_{10}^{22}\text{Ne}$$

4. Answer (4)

$$\text{Potential of conducting hollow sphere} = \frac{KQ}{R}$$

Now, Q = same

$$\Rightarrow V \propto \frac{1}{R} \Rightarrow \text{more the radius less will be the potential.}$$

$$\Rightarrow \text{Hence potential would be more on smaller sphere}$$

5. Answer (1)

We know, velocity of transverse wave

$$v = \sqrt{\frac{T}{\mu}}$$

$$\therefore v_i = \sqrt{\frac{T}{\mu}} \text{ and } v_f = \sqrt{\frac{2T}{\mu}}$$

$$\therefore \frac{v_i}{v_f} = \frac{1}{\sqrt{2}}$$

6. Answer (3)

$$\text{Excess pressure inside the bubble} = \Delta P = \frac{4T}{R}$$

$$P_{\text{in}} = P_{\text{out}} + \frac{4T}{R}$$

as ' R ' increases P_{in} decreases

7. Answer (1)

$$S_{n^{\text{th}}} = u + \frac{1}{2}a(2n-1)$$

$$S_{1^{\text{st}}} = \frac{1}{2}g(2 \times 1 - 1) = \frac{g}{2}$$

$$S_{2^{\text{nd}}} = \frac{1}{2}g(2 \times 2 - 1) = 3\left(\frac{1}{2}g\right)$$

$$S_{3^{\text{rd}}} = \frac{1}{2}g(2 \times 3 - 1) = 5 \times \left(\frac{1}{2}g\right)$$

$$S_{4^{\text{th}}} = \frac{1}{2}g(2 \times 4 - 1) = 7 \times \left(\frac{1}{2}g\right)$$

$$S_{1^{\text{st}}} : S_{2^{\text{nd}}} : S_{3^{\text{rd}}} : S_{4^{\text{th}}}$$

$$= 1 : 3 : 5 : 7$$

8. Answer (4)

We know, magnetic field at centre of solenoid

$$B = \mu_0 \frac{N}{\ell} I = \mu_0 n I \quad \left[n = \frac{N}{\ell} \right]$$

$$= 4\pi \times 10^{-7} \times 100 \times 10^3 \times 1 \quad \left[n = \frac{100}{10^{-3}} \right]$$

$$= 4\pi \times 10^{-2} \text{ T}$$

$$B = 12.56 \times 10^{-2} \text{ T}$$

9. Answer (2)

Slope of $x-t$ curves gives the velocity

$$\Rightarrow \text{Ratio} = \frac{\tan 30^\circ}{\tan 45^\circ} = \frac{1}{\sqrt{3}} = 1 : \sqrt{3}$$

10. Answer (1)

We know,

$$\text{RMS value of A.C. } E_{\text{rms}} = \frac{E_0}{\sqrt{2}}$$

$$E_0 = \sqrt{2} E_{\text{rms}}$$

11. Answer (2)

$$E_n = \frac{E_0}{n^2}, \text{ For first excited state } \Rightarrow n = 2$$

$$\text{For second excited state } \Rightarrow n = 3$$

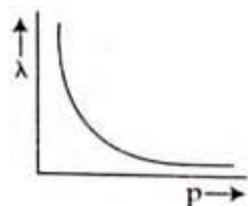
$$\Rightarrow \frac{T_1}{T_2} = \frac{\frac{E_0}{4}}{\frac{E_0}{9}} = \frac{9}{4}$$

12. Answer (2)

de-Broglie wavelength associated with a particle is given by

$$\lambda = \frac{h}{p}$$

$$\lambda \propto \frac{1}{p}$$



13. Answer (3)

$$\text{Plane angle} = \frac{\text{Arc}}{\text{Radius}} = \frac{[L]}{[L]} \longrightarrow \text{Unit} = \text{Radian}$$

$$= [M^0 L^0 T^0]$$

$$\text{Solid angle} = \frac{\text{Area}}{(\text{Radius})^2} \longrightarrow \text{Unit} = \text{Steradian}$$

$$= \frac{L^2}{L^2} = [M^0 L^0 T^0]$$

$\therefore$ Both have units but no dimensions

14. Answer (2)

Potential drops across the p - n junctions will be same if either both junctions are forward biased or both junction are reverse biased.

In figure (a) and (c), both junctions are forward biased therefore both have same potential.

In figure (b) first junction is forward biased and second junction is reverse biased, so both junctions have different potential difference.

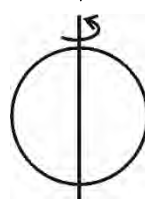
15. Answer (4)



$$I_1 = \frac{MR^2}{2}$$

$$k_1 = \sqrt{\frac{I_1}{M}}$$

$$= \frac{R}{\sqrt{2}}$$



$$I_2 = \frac{MR^2}{4}$$

$$k_2 = \sqrt{\frac{I_2}{M}}$$

$$= \frac{R}{2}$$

$$\frac{k_1}{k_2} = \frac{\frac{R}{\sqrt{2}}}{\frac{R}{2}}$$

$$= \sqrt{2} : 1$$

16. Answer (4)

$$\text{Angular acceleration } \alpha = \frac{\omega_f - \omega_i}{t}$$

$$\omega_f = 3120 \times \frac{2\pi}{60} \text{ rad/s}$$

$$\omega_i = 1200 \times \frac{2\pi}{60} \text{ rad/s}$$

$$\Rightarrow \alpha = \frac{(3120 - 1200)}{16} \times \frac{2\pi}{60} = 4\pi$$

17. Answer (2)

$$\text{Resistance, } R = \rho \frac{L}{A} = \frac{L}{\sigma A}$$

$$\Rightarrow \sigma = \frac{L}{RA}$$

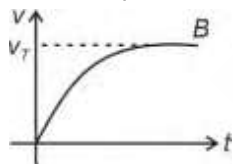
$$\text{Also current density } j = \sigma E = \frac{LE}{RA}$$

$$j = \frac{10 \times 10}{10 \times \pi \left(\frac{10^{-2}}{\sqrt{\pi}} \right)^2} = \frac{100}{10 \times \pi \times \left(\frac{10^{-4}}{\pi} \right)}$$

$$= 10^5 \text{ A/m}^2$$

18. Answer (4)

Initial speed of ball is zero and it finally attains terminal speed



19. Answer (4)

For parallel combination

$$P = \frac{V^2}{R}$$

$$\frac{P_1}{P_2} = \frac{R_2}{R_1}$$

$$\Rightarrow \frac{P_1}{P_2} = \frac{200}{100} = \frac{2}{1}$$

20. Answer (2)

$$1 \text{ eV} = \frac{hc}{\lambda} - \phi_0 \quad \dots(1) \text{ and } 4 \text{ eV} = \frac{3hc}{\lambda} - \phi_0 \quad \dots(2)$$

$$\text{Multiplying eq.(1) by 4} \Rightarrow 4 \text{ eV} = \frac{4hc}{\lambda} - 4\phi_0$$

$$\Rightarrow \frac{4hc}{\lambda} - 4\phi_0 = \frac{3hc}{\lambda} - \phi_0 \Rightarrow \frac{hc}{\lambda} = 3\phi_0$$

$$\Rightarrow \phi_0 = \frac{hc}{3\lambda}$$

21. Answer (2)

$$\beta = \frac{\lambda D}{d}$$

Let length of segment of screen = l

$$\Rightarrow l = 8\beta_1 = \frac{8\lambda_1 D}{d} \quad \dots(1)$$

$$\text{and } l = n\beta_2 = \frac{n\lambda_2 D}{d} \quad \dots(2)$$

from (1) and (2)

$$8\lambda_1 = n\lambda_2$$

$$8(600 \text{ nm}) = n(400 \text{ nm})$$

$$n = 12$$

22. Answer (2)

$$v = \frac{1}{\sqrt{\epsilon_m \mu_m}}$$

$$v = \frac{1}{\sqrt{\epsilon_0 \epsilon_r \mu_0 \mu_r}}$$

$$\text{Since } c = \frac{1}{\sqrt{\epsilon_0 \mu_0}}$$

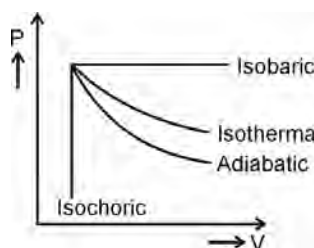
$$\Rightarrow v = \frac{c}{\sqrt{\epsilon_r \mu_r}}$$

23. Answer (4)

$$\left(\frac{dP}{dV} \right)_{\text{adiabatic}} = -\gamma P$$

$$\left(\frac{dP}{dV} \right)_{\text{isothermal}} = -P$$

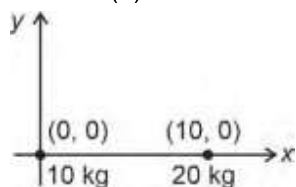
$$\left(\frac{dP}{dV} \right)_{\text{adiabatic}} > \left(\frac{dP}{dV} \right)_{\text{isothermal}}$$



24. Answer (1)

As the temperature increases the resistivity of the conductor increases hence the electrical resistance increases. However for semiconductor the resistivity decreases with the temperature. Hence electrical resistance of semiconductor decreases.

25. Answer (4)



$$X_{cm} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$= \frac{10 \times 0 + 20 \times 10}{10 + 20}$$

$$= \frac{200}{30} = \frac{20}{3} \text{ m}$$

26. Answer (1)

In half wave rectifier, the output frequency is same as that of input frequency.

27. Answer (1)

According to Biot-Savart's law $d\vec{B} = \frac{\mu_0}{4\pi} \frac{Id\vec{l} \times \vec{r}}{r^3}$

which is applicable for infinitesimal element. It is analogous to Coulomb's law, where $Id\vec{l}$ is vector source and electric field is produced by scalar source q .

Here statement I is correct and statement II is incorrect.

28. Answer (3)

Energy = Power $\times$ time

$$E = 100 \times 10^3 \times 3600$$

$$= 36 \times 10^7 \text{ J}$$

29. Answer (4)

$$F = mE_G$$

$$3 = \frac{60}{1000} E_G$$

$$E_G = 50 \text{ N/kg}$$

30. Answer (1)

Power of lens is given by

$$P = \frac{1}{f(m)}$$

$$\frac{1}{f} = (\mu - 1) \left(\frac{1}{R_1} - \frac{1}{R_2} \right)$$

$$\frac{1}{f} = \left\{ \frac{3}{2} - 1 \right\} \left(\frac{1}{20} + \frac{1}{20} \right)$$

$$f = 20 \text{ cm}$$

$$P = \frac{1}{20 \times 10^{-2}}$$

$$= 5 \text{ D}$$

31. Answer (1)

Momentum of the system would remain conserved.

Initial momentum = 0

Final momentum should also be zero.

Let masses be $2m$, $2m$ and m

Momentum along x-direction = $2mv\hat{i}$

Momentum along y-direction = $2mv\hat{j}$

$$\text{Net momentum} = \sqrt{(2mv)^2 + (2mv)^2} = \sqrt{2} \cdot 2mv$$

$$\text{Now, } 2\sqrt{2}mv = mv'$$

$$v' = 2\sqrt{2}v$$

32. Answer (1)

Dimensional formula of magnetic permeability is $[MLT^{-2}A^{-2}]$

33. Answer (1)

$$dV = -\vec{E} \cdot d\vec{r}$$

$$dV = -Edr \cos \theta$$

For equipotential surface,

$$dV = 0$$

$$\cos \theta = 0$$

$$\Rightarrow \theta = 90^\circ$$

34. Answer (4)



Magnetic flux $(\phi_B) = \vec{B} \cdot \vec{A}$

$\vec{B}$ and $\vec{A}$ are in same direction, therefore

$$\phi_B = B \cdot A = 0.5 \times 1^2$$

$$= 0.5 \text{ Wb}$$

35. Answer (1)

Given $i = 60^\circ$ and $\mu = \sqrt{3}$

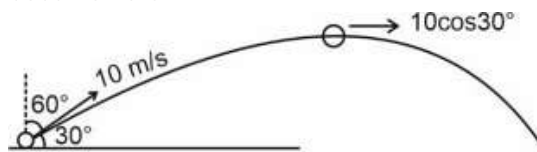
$\Rightarrow$ Here, angle of incidence $\Rightarrow i = \tan^{-1}(\mu)$

Hence, reflected and refracted rays would be perpendicular to each other.

SECTION - B

36. Answer (4)

At highest point vertical component of velocity become zero.



At highest point speed of object = $10 \cos 30^\circ$

37. Answer (4)

$$\omega L = \frac{1}{\omega C}$$

$$\omega = \frac{1}{\sqrt{LC}} = \frac{1}{\sqrt{10 \times 10 \times 10^{-6}}}$$

$$\omega = 100$$

$$\omega = 2\pi f \Rightarrow f = \frac{\omega}{2\pi}$$

$$v_0 = f_0 = \frac{100}{2\pi} = \frac{50}{\pi} \text{ Hz}, \quad \omega = 100$$

$$v = f = \frac{100}{2\pi} = \frac{50}{\pi} \text{ Hz}$$

38. Answer (2)

$$\text{Area} = \text{Length} \times \text{Breadth}$$

$$= 55.3 \times 25 \text{ m}^2 = 1382.5 \text{ m}^2$$

$$= 14 \times 10^2 \text{ m}^2 \text{ (Rounding off of two significant figures)}$$

39. Answer (3)

$$T = 2\pi\sqrt{\frac{L}{g}}$$

Let n_1 and n_2 be integer.

$$n_1 T_1 = n_2 T_2$$

$$2\pi n_1 \sqrt{\frac{1.21}{g}} = 2\pi n_2 \sqrt{\frac{1.00}{g}}$$

$$\Rightarrow \frac{n_2}{n_1} = \frac{11}{10}$$

$\therefore$ After completion of 11th oscillation of shorter pendulum, it will be in phase with longer pendulum.

40. Answer (1)

It is true that stretching of spring is determined by shear modulus of the spring as when coil spring is stretched neither its length nor its volume changes, there is only change in its shape.

Tensile strength of steel is more than that of copper.

Hence Assertion is true and reason is false.

41. Answer (4)

We know, a wheatstone bridge is said to be most precise when it is most sensitive. This can be done by making ratio arms equal. Thus (4) is correct option.

42. Answer (4)

For $R \gg L$, arrangement is an electric dipole

$$E = \frac{2p}{4\pi\epsilon_0 R^3}; \text{ where } p = qL$$

$$E \propto \frac{1}{R^3}$$

43. Answer (1)

Radius of nuclei with mass number A varies as

$$R = R_0 A^{1/3}$$

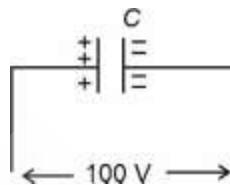
$$\frac{R_1}{R_2} = \left(\frac{125}{64}\right)^{1/3} = \frac{5}{4} = 5 : 4$$

44. Answer (1)

$$q_1 = CV$$

$$= 900 \times 10^{-12} \times 100$$

$$= 9 \times 10^{-8} \text{ C}$$



$$V = \frac{C_1 V_1 + C_2 V_2}{C_1 + C_2}$$

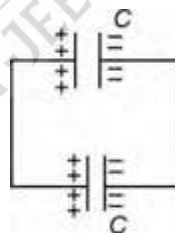
$$= \frac{9 \times 10^{-8} + 0}{1800 \times 10^{-12}} = \frac{100}{2} = 50 \text{ V}$$

$$U = \frac{1}{2}(C_1 + C_2)V^2$$

$$= \frac{1}{2} \times 1800 \times 10^{-12} \times 50 \times 50$$

$$= 225 \times 10^{-8}$$

$$U = 2.25 \times 10^{-6} \text{ J}$$



45. Answer (1)

$$C = (\overline{A \cdot B}) \cdot (\overline{\overline{A} \cdot \overline{B}})$$

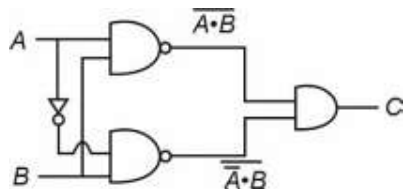
$$\Rightarrow C = \overline{A \cdot B} + \overline{\overline{A} \cdot \overline{B}}$$

$$\Rightarrow C = \overline{(A + \overline{A})B}$$

$$\Rightarrow C = \overline{B}$$

The truth table would be

A	B	C
0	0	1
0	1	0
1	0	1
1	1	0



46. Answer (2)

From ideal gas equation

$$PV = nRT$$

$$\left[n = \frac{\text{mass of water}}{\text{mol. wt.}} = \frac{4.5 \times 10^3}{18} \right]$$

$$V = \frac{nRT}{P}$$

At. STP $\Rightarrow T = 273 \text{ K}$

$$P = 10^5 \text{ N/m}^2$$

$$V = \frac{4.5 \times 10^3}{18} \times \frac{8.3 \times 273}{10^5} = 5.66 \text{ m}^3$$

47. Answer (4)

$$(a) [G] = \frac{Fr^2}{m_1 m_2}$$

$$[G] = \frac{Fr^2}{m_1 m_2} = \frac{[MLT^{-2}] L^2}{[MM]} = [M^{-1} L^3 T^{-2}]$$

(b) Gravitational potential energy = $[ML^2 T^{-2}]$

$$(c) \text{Gravitational potential} = \frac{PE}{m} = [L^2 T^{-2}]$$

$$(d) \text{Gravitational field intensity} = \frac{F}{m} = [LT^{-2}]$$

48. Answer (1)

For solid wire

Inside point

$$B = \frac{\mu_0 I r^2}{R^2 \times 2\pi r}$$

$$= \frac{\mu_0 I r}{R^2 \times 2\pi}$$

$$B \propto r$$

Outside point

$$B = \frac{\mu_0 I}{2\pi r}$$

$$B \propto \frac{1}{r}$$

49. Answer (1)

$$\phi_B = NBA \cos \omega t$$

$$\varepsilon = \frac{-d\phi_B}{dt} = -NBA\omega(-\sin \omega t)$$

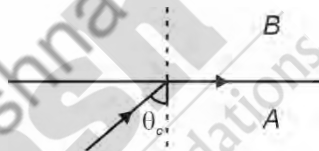
$$\varepsilon = NBA\omega \sin \omega t$$

$$i_{\max} = \frac{\varepsilon_{\max}}{R} = \frac{NBA\omega}{R}$$

$$= \frac{1000 \times 2 \times 10^{-5} \times \pi (10)^2 \times 2}{12.56}$$

$$= 1 \text{ A}$$

50. Answer (4)



$$\mu_A = \frac{3 \times 10^8}{1.5 \times 10^8} = 2$$

$$\mu_B = \frac{3 \times 10^8}{2 \times 10^8} = 1.5$$

For TIR, ray of light should travel from denser to rarer medium

$$\mu_A \sin \theta_c = \mu_B \sin 90^\circ$$

$$2 \sin \theta_c = 1.5 \sin 90^\circ$$

$$\sin \theta_c = 0.75$$

$$\theta_c = \sin^{-1}(0.75)$$

CHEMISTRY

SECTION - A

51. Answer (1)

Diamond : • sp^3 hybridised carbon atom

- Covalent solid
- 3-D structure
- Cannot be used as dry lubricant

Graphite : • sp^2 hybridised carbon atom

- Covalent solid
- 3-D structure
- Used as dry lubricant

52. Answer (2)

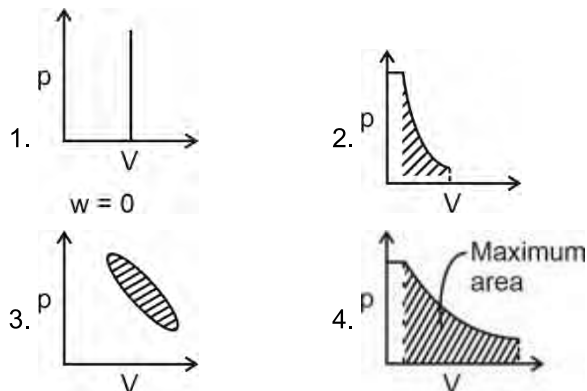
 ΔU for isothermal process = 0 ΔU for ideal gas depends only on temperatureHence, for isothermal process $s = \Delta T = 0$

$$\Delta U = 0$$

53. Answer (4)

Work done under any thermodynamic process can be determined by area under the 'p-V' graph.

As it can be observed maximum area is covered in option '4'.



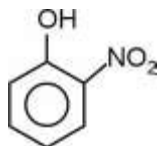
54. Answer (1)

Kjeldahl method is not applicable to compounds containing nitrogen in nitro group, azo groups and nitrogen present in the ring (e.g., pyridine) as nitrogen of these compounds does not change to ammonium sulphate under these conditions.

55. Answer (2)

Different gases have different K_H value, so the gas which has high K_H value. Has low solubility into the solvent.

56. Answer (1)



- Nitro group has electron withdrawing tendency. It can withdraw electrons both by $-I$ effect and $-R$ effect. Thus the acidic strength of monosubstituted nitrophenol is higher than phenol.
- Nitro group present at o - and p -positions will have strong $-R$ effect while nitro group present at m -position will influence only $-I$ effect hence acidity of o/p isomer will be more meta isomer.

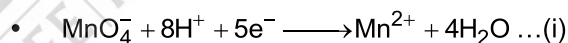
57. Answer (2)

(Quantum Number)	(Orbital)
$n = 2, l = 1$	$2p$
$n = 3, l = 2$	$3d$
$n = 3, l = 0$	$3s$
$n = 2, l = 0$	$2s$

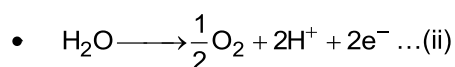
58. Answer (1)

Example	Dipole moment, μ (B)
BF_3	0
CCl_4	0
CH_4	0
	0
NF_3	0.23
NH_3	1.47
H_2O	1.85
CO_2	0
	1.68

59. Answer (3)

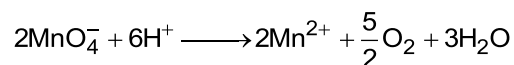


$$E_{\text{MnO}_4^-/\text{Mn}^{2+}}^\circ = -E_{\text{Mn}^{2+}/\text{MnO}_4^-}^\circ = 1.51 \text{ V}$$



$$E_{\text{O}_2/\text{H}_2\text{O}}^\circ = 1.223 \text{ V}$$

Using $2 \times (i) + 5 \times (ii)$, net cell reactions is



$$E_{\text{cell}}^\circ = E_{\text{C}}^\circ - E_{\text{A}}^\circ = E_{\text{MnO}_4^-/\text{Mn}^{2+}}^\circ - E_{\text{O}_2/\text{H}_2\text{O}}^\circ \\ = 1.51 - 1.223 = 0.287 \text{ V}$$

Since $E_{\text{cell}}^\circ > 0$, therefore net cell reaction is spontaneous and so MnO_4^- liberate O_2 from H_2O in presence of an acid.

70. Answer (4)

Compound	Boiling point (K)
H ₂ O	373
H ₂ S	213
H ₂ Se	232
H ₂ Te	269

- The boiling points of these hydrides not exactly increases with increase in molar mass.
- H₂O has maximum boiling point due to intermolecular hydrogen bonding.

71. Answer (4)

Molality is the moles of solute dissolved per kg of solvent therefore 500 g, 1 molal solution contains 0.5 mol of solute, as

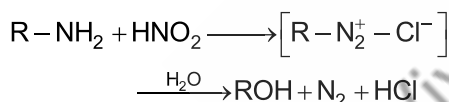
$$m = \frac{\text{Moles of solute}}{\text{Mass of solvent (in kg)}}$$

$$1 = \frac{0.5}{\text{Mass of solvent (in kg)}}$$

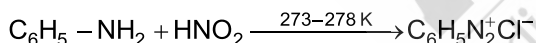
$$\therefore \text{Mass of solvent (in kg)} = 0.5 \\ = 500 \text{ g}$$

72. Answer (1)

- Primary aliphatic amines react with HNO₂ and give unstable diazonium salt which turns into alcohol



- Primary aromatic amines reacts with HNO₂ and give stable diazonium salt which are stable at 273 to 278 K.



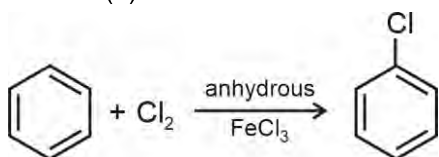
73. Answer (4)

Electronic configuration of Gadolinium

Gd :— [Xe] 4f⁷ 5d¹ 6s²

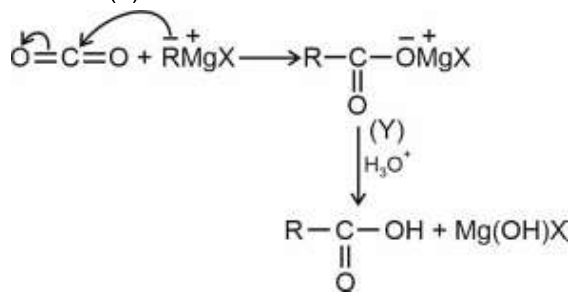
In case of 3rd ionisation enthalpy electron will be removed from 5d and resultant configuration will be [Xe]4f⁷ that is stable electronic configuration as it will have high exchange energy, hence less energy will be required to remove 3rd electron.

74. Answer (3)



Benzene reacts with chlorine in presence of anhydrous FeCl₃ to give chlorobenzene

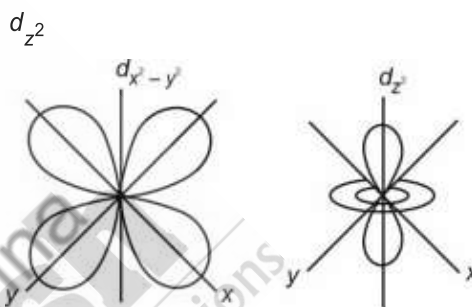
75. Answer (3)



Here Y is RCOO-Mg⁺X

76. Answer (2)

- In an atom, all the five 3d orbitals are equal in energy in free state i.e., degenerate.
- The shape of $d_{x^2-y^2}$ is different than shape of



- The size of orbital depends on principal quantum number 'n' therefore all the five 3d orbitals are different in size when compared to the respective 4d orbitals.

- Shape of orbitals depends on azimuthal quantum number 'l' therefore shapes of 4d orbitals are similar to the respective 3d orbitals.

77. Answer (3)

IUPAC name of element : 119 : Ununennium

78. Answer (3)

- The boiling points of aldehydes and ketones are higher than hydrocarbons of comparable molecular masses due to stronger molecular association in aldehydes and ketones arising out of the dipole - dipole interaction.
- Alcohols involved intermolecular hydrogen bonding, because of which the boiling point of aldehydes and ketones are lower than the alcohols of similar molecular masses.

79. Answer (4)

Let m gram mass of CaCO₃ is required

$$\text{Pure CaCO}_3 \text{ in m gram} = \frac{95}{100} \times m$$

$$\text{Moles of CaCO}_3 = \frac{95}{100} \times \frac{m}{100}$$

Moles of HCl required = 2 × moles of CaCO₃

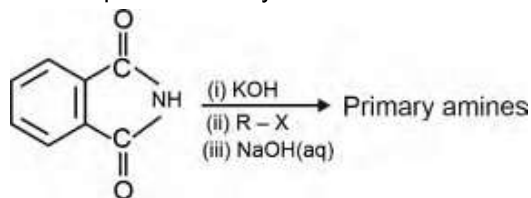
$$= 2 \times \frac{95}{100} \times \frac{m}{100}$$

$$2 \times \frac{95}{100} \times \frac{m}{100} = \frac{50}{1000} \times 0.5$$

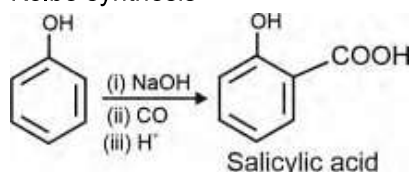
$$m = 1.315 \text{ g} \approx 1.32 \text{ g}$$

80. Answer (1)

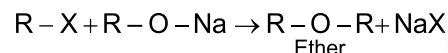
Gabriel phthalimide synthesis



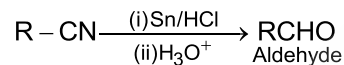
Kolbe synthesis



Williamson reaction



Stephen's synthesis



81. Answer (1)

- For zero order reaction

$$r = k[A]^0$$

$$r = k \text{ (constant)}$$

hence, 'y' as 'rate' and 'x' as concentration will give desired graph.

- For first order reaction

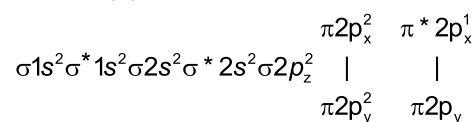
$$t_{1/2} = \frac{0.693}{k} \text{ (constant)}$$

hence, 'y' as 't_{1/2}' and 'x' as concentration will give desired graph.

82. Answer (1)

In general, interhalogen compounds are more reactive than halogens (except fluorine). This is because X - X' bond in interhalogens is weaker than X - X bond in halogens excepts F - F bond.

83. Answer (2)



Due to one unpaired electron in $\pi^* 2p$ molecular orbital, O₂⁺ is a paramagnetic ion.

84. Answer (2)

For a reaction to be spontaneous, E_{cell}^o must be positive.

- For, FeSO₄(aq) + Zn(s) → ZnSO₄(aq) + Fe(s)

$$\begin{aligned} E_{\text{cell}}^o &= E_{\text{cathode}}^o - E_{\text{anode}}^o \\ &= -0.44 \text{ V} - (-0.76 \text{ V}) \\ &= 0.32 \text{ V} \end{aligned}$$

- For, 2CuSO₄(aq) + 2Ag(s) → 2Cu(s) + Ag₂SO₄(aq)

$$E_{\text{cell}}^o = 0.34 \text{ V} - 0.80 \text{ V} = -0.46 \text{ V}$$

- For, CuSO₄(aq) + Zn(s) → ZnSO₄(aq) + Cu(s)

$$\begin{aligned} E_{\text{cell}}^o &= 0.34 \text{ V} - (-0.76 \text{ V}) \\ &= 1.1 \text{ V} \end{aligned}$$

- For, CuSO₄(aq) + Fe(s) → FeSO₄(aq) + Cu(s)

$$\begin{aligned} E_{\text{cell}}^o &= 0.80 \text{ V} - (-0.44 \text{ V}) \\ &= 1.24 \text{ V} \end{aligned}$$

85. Answer (1)

The stereoisomers related to each other as non-superimposable mirror image are called enantiomers.

SECTION-B

86. Answer (1)

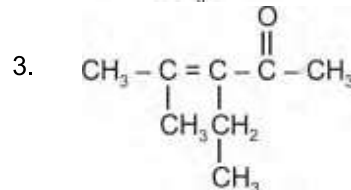
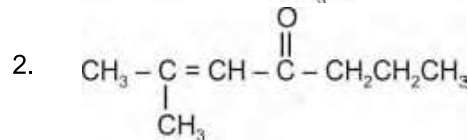
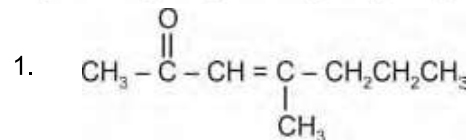
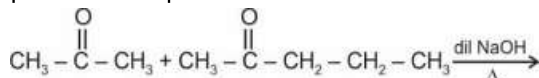
Statement I is correct but statement II is incorrect.

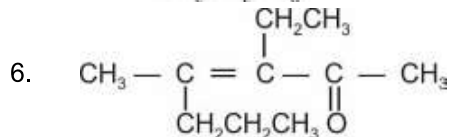
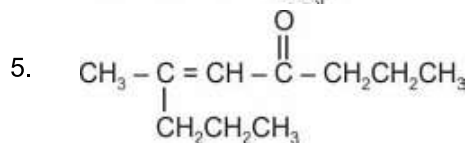
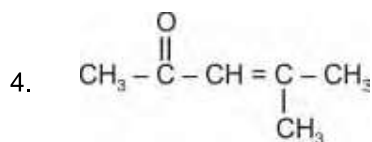
Compound	Formula	pK _a
Phenol	C ₆ H ₅ - OH	10.0
o-Cresol	o - CH ₃ - C ₆ H ₄ - OH	10.2
m-Cresol	m - CH ₃ - C ₆ H ₄ - OH	10.1
p-Cresol	p - CH ₃ - C ₆ H ₄ - OH	10.2

87. Answer (4)

Cross Aldol condensation reaction:

Both reactants contain α-Hydrogens, so multiple products are possible which are as follows:



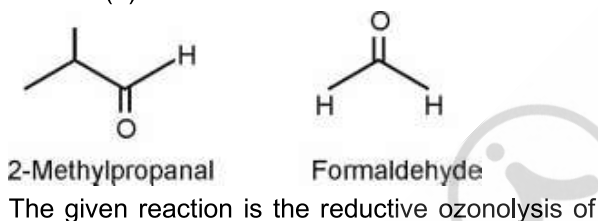


$\Rightarrow$ (4) is not possible.

88. Answer (3)

Except $\text{NH}_2 - \text{NH}_2$, all the remaining three compounds contain 'C' and 'N', needed for the formation of NaCN and thus give blue colour.

89. Answer (3)



an alkene. The alkene will be 

90. Answer (3)

$$r_n \propto \frac{n^2}{7}$$

$$\frac{r_3(\text{Li}^{2+})}{r_2(\text{He}^+)} = \frac{(n_3)^2}{Z(\text{Li}^{2+})} \times \frac{Z(\text{He}^+)}{(n_2)^2}$$

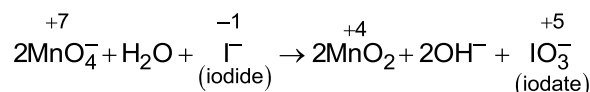
$$\frac{r_3(\text{Li}^{2+})}{105.8} = \frac{(3)^2}{3} \times \frac{2}{(2)^2}$$

$$= 105.8 \times \frac{3}{2}$$

$$r_3(\text{Li}^{2+}) = 158.7 \text{ pm}$$

91. Answer (3)

In neutral or faintly alkaline solution.



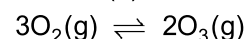
Manganese (Mn) oxidation state change from +7 to +4.

92. Answer (1)

Primary, secondary and tertiary alcohols can be differentiated by their reaction with (HCl + anhydrous ZnCl_2) Lucas reagent

- 3° alcohol $\xrightarrow{\text{ZnCl}_2 + \text{HCl}}$ Immediate turbidity at room temperature
- 2° alcohol $\xrightarrow{\text{ZnCl}_2 + \text{HCl}}$ Turbidity after 5 minutes at room temperature
- 1° alcohol $\xrightarrow{\text{ZnCl}_2 + \text{HCl}}$ Do not gives turbidity at room temperature

93. Answer (3)



$$K_C = \frac{[O_3]^2}{[O_2]^3}$$

$$[O_3]^2 = K_G [O_2]^3 = 3 \times 10^{-59} \times (0.04)^3$$

$$[\text{O}_3]^2 = 1.9 \times 10^{-63} = 19 \times 10^{-64}$$

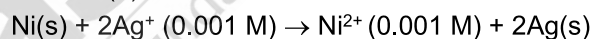
$$[\text{O}_3] = 4.38 \times 10^{-32}$$

Concentration of O_3 at equilibrium
 $= 4.38 \times 10^{-32} \text{ M}$

94. Answer (4)

In strong acid strong base titration like HCl with NaOH, equivalence point lies in the pH range 3-11. Thus, methyl orange, methyl red and phenolphthalein will be suitable indicators.

95. Answer (1)



$$E_{\text{cell}}^{\circ} = 1.05 \text{ V}$$

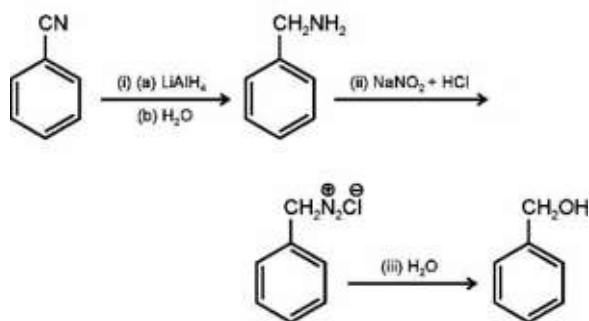
$$E_{\text{cell}} = E_{\text{cell}}^{\circ} - \frac{0.059}{n} \log \frac{[\text{Ni}^{2+}]}{[\text{Ag}^+]^2}$$

$$= 1.05 - \frac{0.059}{2} \log \frac{(10^{-3})}{(10^{-3})^2}$$

$$\Rightarrow 1.05 - \frac{0.059}{2} \log(10)^3$$

$$\begin{aligned} &\Rightarrow 1.05 - 0.0295 \times 3 \\ &= 1.05 - 0.0885 \\ &= 0.96 \text{ V} \end{aligned}$$

96. Answer (2)



97. Answer (4)

For first order reaction,

$$k = \frac{2.303}{t} \log \frac{[A_0]}{[A]}; \quad \text{where } A_0 \text{ is the initial}$$

concentration of reactant A.

$$A_0 = 0.1 \text{ M}$$

$$A = 0.001 \text{ M}$$

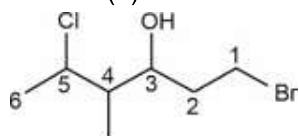
$$t = 5 \text{ minute}$$

$$k = \frac{2.303}{5} \log \frac{0.1}{0.001} = \frac{2.303}{5} \log 10^2$$

$$= \frac{2.303}{5} \times 2$$

$$k = 0.9212 \text{ min}^{-1}$$

98. Answer (3)



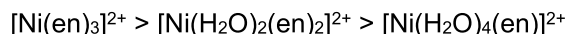
1-bromo-5-chloro-4-methylhexan-3-ol

99. Answer (1)

Stronger the field strength of ligand, higher will be the energy absorbed by the complex.

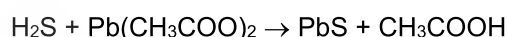
⇒ 'en' has a stronger field strength than 'H₂O' according to spectrochemical series

∴ Correct order of energy absorbed will be:



i.e. (C) > (A) > (B)

100. Answer (3)



(Turns lead acetate paper black)

Possible Anion = (S²⁻)

BOTANY

SECTION - A

101. Answer (1)

Chitin forms the exoskeleton in arthropods and is found in the fungal cell wall as well. N-acetyl glucosamine is the monomeric unit of chitin. Cellulose is a polysaccharide. Cutin is a derived lipid.

102. Answer (2)

Plants follow different pathways in response to environment or phases of life to form different kinds of structures. This ability is called plasticity e.g. heterophylly in cotton, coriander and larkspur. In such plants, leaves of juvenile plant are different in a shape from those in mature plants.

Maize does not show plasticity.

103. Answer (4)

Chemiosmosis requires a membrane, a proton pump, a proton gradient and ATP synthase. Energy is used to pump protons across a membrane to create a gradient or a high concentration of protons within the thylakoid lumen.

The NADP reductase enzyme is located on the stroma side of the membrane. Along with the electrons that come from the acceptor of electrons of PS I, protons are necessary for reduction of NADP⁺ to NADPH + H⁺.

The process does not involve breaking of electron gradient.

104. Answer (3)

Gregor J. Mendel, conducted hybridisation experiments on garden peas and selected 14 true breeding pea plant varieties (seven contrasting traits). Contrasting traits studied were smooth or wrinkled seeds, yellow or green seeds, inflated or constricted pods, green or yellow pods, tall or dwarf plants, violet or white flowers and axial or terminal flower positions.

105. Answer (1)

Hydrocolloids are water holding substances for eg. carrageen obtained from red algae (Rhodophyceae).

106. Answer (1)

Decomposition is the process by which decomposers breakdown complex organic matter into inorganic substances.

The rate of decomposition is controlled by chemical composition of detritus and climatic factors. Decomposition is slower if detritus is rich in lignin and chitin and quicker, if detritus is rich in nitrogen and water soluble substances like sugars.

107. Answer (4)

Dinoflagellates have cell wall of stiff cellulose plates on the outer surface (e.g. *Gonyaulax*)

108. Answer (1)

Axillary buds of stems get modified into woody, straight and pointed thorns. Thorns are found in many plants such as *Citrus* and *Bougainvillea*.

109. Answer (4)

When a flower can be divided into two similar halves only in one particular vertical plane, it is zygomorphic for e.g. pea, gulmohar, bean, *Cassia*. Mustard, *Datura* and Chilli show actinomorphic flowers.

110. Answer (4)

The first action spectrum obtained by T.W. Englemann roughly resembles the absorption spectra of chlorophyll *a* and *b*.

111. Answer (2)

Among the animals, insects, particularly bees are the dominant biotic pollinating agents.

112. Answer (3)

The primary CO₂ acceptor is a 3-carbon molecule, phosphoenol pyruvate (PEP) and is present in the mesophyll cells.

Mesophyll cells of C₄ plants lack RuBisCO enzyme.

113. Answer (4)

In-situ conservation means on site conservation i.e. when we conserve and protect the whole ecosystem, its biodiversity at all levels is protected. National parks are type of *in-situ* conservation.

Whereas, micropropagation, cryopreservation and *in-vitro* fertilisation are methods of *ex-situ* conservation.

114. Answer (2)

Less than seven percent of the energy in glucose is released during lactic acid fermentation and not all of it is trapped as high energy bonds of ATP.

115. Answer (4)

- Vexillary aestivation and diadelphous stamens are the characteristic features of family Fabaceae.
- *Pisum sativum* (garden pea) belongs to family Fabaceae.
- *Allium cepa* (onion) and *Colchicum autumnale* (colchicine) belong to family Liliaceae.
- *Solanum nigrum* belongs to Solanaceae.

116. Answer (4)

Heterochromatin is transcriptionally inactive. A typical nucleosome contains 200 bp of DNA helix. Euchromatin is the loosely packed chromatin region.

The negatively charged DNA is wrapped around the positively charged histone octamer to form a structure called nucleosome. Histones are rich in basic amino acid residues lysine and arginine.

117. Answer (4)

Ethylene is a gaseous plant hormone. It induces development of adventitious roots on various types of cutting. It promotes the development of lateral roots and growth of root hairs. Cytokinin helps to overcome the apical dominance.

Auxin is used to kill dicot weeds. Gibberellin speeds up the malting process

118. Answer (1)

Grasshopper is an example of XO type of sex determination in which the males have only one X-chromosome besides the autosomes, whereas females have a pair of X-chromosomes.

119. Answer (4)

The plant growth regulators (PGRs) are small molecules of diverse chemical composition. They could be

- Adenine derivatives (N₆-furfurylamino purine, kinetin)
- Carotenoids derivatives (Absciscic acid, ABA)
- Terpenoids derivatives (Gibberellic acid, GA₃)
- Gaseous PGR (Ethylene, C₂H₄)

120. Answer (4)

Species that are confined to a particular region and not found anywhere else is called endemism.

121. Answer (3)

Cleistogamous flowers does not open at all. In such flowers autogamy occurs. Lack of cross pollination is a disadvantage of cleistogamy.

122. Answer (4)

Ulothrix is a member of Chlorophyceae (green algae), with reserve food material, starch.

Mannitol is stored food material of Phaeophyceae (brown algae).

123. Answer (1)

Pairing of homologous chromosomes occurs during prophase I of meiosis.

Coiling and condensation of chromatids, spindle fibres attachment to the kinetochores and movement of centrioles towards opposite poles occur in both mitosis and meiosis.

124. Answer (1)

Ethylene increases the number of female flowers and fruits in certain plants such as cucumber. Gibberellins are used to increase the size of fruits in some plants.

125. Answer (1)

Bright orange colored bands of DNA can be observed in the gel when EtBr (Ethidium bromide) treated DNA is exposed to UV light.

126. Answer (1)

Pachytene stage of meiosis is characterised by the appearance of recombination nodules, the sites at which crossing over occurs between non sister chromatids of homologous chromosomes.

127. Answer (3)

If fatty acid were to be respired they would first be degraded to acetyl CoA and enter the respiratory pathway. Glycerol would enter the pathway after being converted to 3-phosphoglycerdehyde (PGAL)

128. Answer (3)

When the small subunit of ribosome encounters an mRNA, the process of translation of the mRNA to protein begins. This process is followed by the binding of bigger/larger subunit.

t-RNA is activated by the addition of amino acid prior to the attachment of ribosome, in the first phase.

129. Answer (1)

During glycolysis, total 4 ATPs are produced from one glucose molecule with a net gain of 2 ATPs.

130. Answer (1)

One of the species in predation gains benefit on the expense of the other. Predators help in maintaining species diversity in a community, by reducing the intensity of competition among competing prey species. If a predator is too efficient and overexploits its prey, then the prey might become extinct.

131. Answer (1)

Habitat loss and fragmentation, over exploitation, alien species invasion and co-extinction are causes for biodiversity loss.

132. Answer (3)

In open vascular bundle, cambium is present in between xylem and phloem. In monocotyledonous root, usually there are more than six xylem bundles present.

133. Answer (1)

Polymorphism in DNA sequence is the basis of genetic mapping of human genome as well as of DNA fingerprinting.

134. Answer (3)

In old trees, the greater part of secondary xylem is dark brown due to deposition of organic compounds like tannins, resins, oils, gums, aromatic substances and essential oils in the central or innermost layers of the stem. These substances make it hard, durable and resistant to the attacks of micro-organisms and insects.

135. Answer (4)

Both the statements are correct but the given reason is not the correct explanation of the assertion. Polymerase chain reaction is used in DNA amplification.

Ampicillin resistance gene is a selectable marker that helps to check transformation by selection of transformants.

SECTION - B

136. Answer (4)

Spring wood is also called early wood. It is lighter in colour and has a lower density. The vessels are produced with the wider lumens to transport more water to meet the requirement by increased transpiring surface in spring season.

The spring and autumn wood appear as alternate concentric rings of light and dark colour forming annual rings.

137. Answer (3)

In metacentric chromosome, centromere is in the middle of the chromosomes. Acrocentric chromosome has centromere close to the end of the chromosome. In submetacentric chromosome, centromere is slightly away from the middle of the chromosome. Telocentric chromosome has terminal centromere.

138. Answer (1)

The given figure is of a false fruit. False fruit develops from other floral parts and thalamus forms the main edible part.

139. Answer (4)

A palindromic DNA sequence is a DNA sequence of base pairs that reads the same on the two strands when the orientation of reading is kept the same. Out of the four given options, option (4) is the only palindromic sequence.

5'GAATTC3'

3'CTTAAG5'

140. Answer (4)

The large cells around the vascular bundles of C₄ plants form bundle sheath. These cells have large number of chloroplasts to perform Calvin cycle.

141. Answer (3)

In secondary treatment or biological treatment, the primary effluent is passed into large aeration tanks where it is constantly agitated mechanically and air is pumped into it. Once the BOD of sewage or waste water is reduced significantly the effluent is then passed into a settling tank.

142. Answer (4)

Spirogyra is an alga. It shows haplontic life-cycle. Fern is a pteridophyte. The dominant phase of life-cycle is diploid sporophyte. Its gametophyte is called prothallus.

Funaria is a bryophyte. Its gametophyte is a leafy stage.

Cycas is a gymnosperm. The main plant body in gymnosperm is sporophyte. They have highly reduced gametophyte stage.

143. Answer (3)

When oxygen is inadequate for cellular respiration pyruvate acid is reduced to lactic acid by lactate dehydrogenase. The reducing agent is $\text{NADH} + \text{H}^+$ which is deoxidised to NAD^+ .

144. Answer (1)

Phosphorus cycle is a sedimentary cycle. Reservoir pool of phosphorus in ecosystem is the earth's crust or lithosphere. Weathering of rocks accelerate phosphorus cycle.

145. Answer (3)

Sequencing the whole set of genome that contained all the coding and non-coding sequences and later assigning different regions in the sequence with functions is called sequence annotation.

146. Answer (4)

Haemophilia is a X-linked recessive disorder. Thalassaemia is an autosomal recessive disorder. Sickle cell anaemia is an autosomal recessive disorder.

Myotonic dystrophy is an autosomal dominant disorder i.e. it occurs due to the presence of autosomal linked dominant trait.

147. Answer (1)

Closely located genes do not show independent assortment. Mendel's law of independent assortment holds good for those genes which are located on different chromosomes.

148. Answer (3)

In predation, one species is benefitted whereas the other is harmed. It is (+, -) type of population interaction.

149. Answer (1)

Statements (c), (d) and (e) are correct as oils have lower melting point and hence they remain as oil in winters. Lipids are generally insoluble in water but soluble in some organic solvents.

Statements (a) and (b) are incorrect. Lecithin is a type of phospholipid found in the plasma membrane. Saturated fatty acids are without double bonds.

150. Answer (4)

The source of the complementary RNA for RNAi could be mobile genetic elements (transposons) that replicate via an RNA intermediate.

Option (1) is incorrect as autoradiography usually follows hybridisation.

Option (3) is incorrect because polymerase chain reaction is used to make multiple copies of the DNA sample and does not need transposons.

Option (2) is incorrect because transposons are not required during gene sequencing.

ZOOLOGY

SECTION - A

151. Answer (3)

When we conserve and protect the whole ecosystem, its biodiversity at all levels is protected. This is *in-situ* or on site conservation strategy.

152. Answer (2)

The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part; whereas the alveoli and their ducts form the respiratory or exchange part of the respiratory system.

The conducting part transports the atmospheric air to the alveoli, clears it from foreign particles, humidifies and also brings the air to body

temperature. Exchange part is the site of actual diffusion of O_2/CO_2 between blood and atmospheric air.

153. Answer (4)

Conidia are the asexual reproductive structures produced in *Penicillium*.

Gemmules are produced in sponge

Buds are produced in *Hydra*

Zoospores are produced in *Chlamydomonas*

154. Answer (1)

Mycoplasma are the smallest cells and are only $0.3 \mu\text{m}$ in length. So it can pass through less than $1 \mu\text{m}$ filter size.

Mycoplasma lack cell wall.

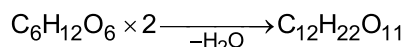
155. Answer (4)

Tegmina or forewings (the first pair of wings) in cockroach arises from the mesothorax.

No wing arises from prothorax and hindwings arise from the metathorax.

156. Answer (1)

Maltose is a disaccharide formed by the dehydration process *i.e.*, synthesis occurs by the elimination of one water molecule to form a glycosidic bond in between two glucose molecules. So, its molecular formula will be



157. Answer (1)

Osteoporosis occurs due to decreased levels of estrogen.

Osteoporosis is an age-related disorder characterised by decreased bone mass hence, the chances of fractures increase.

158. Answer (4)

In directional natural selection, more individuals acquire value other than the mean character value.

In disruptive change, more individuals acquire peripheral character value at both ends of the distribution curve.

There is no random change in natural selection.

Natural selection leads to stabilisation when more individuals acquire mean character value.

159. Answer (3)

Each restriction endonuclease recognises a specific palindromic nucleotide sequence in the DNA. It will bind to the DNA and cut each of the two strands of double helix at specific points.

Restriction enzymes cut the strand of DNA a little away from the centre of the palindrome site; but between the same two bases on the opposite strands.

160. Answer (2)

Neuroglia are the part of neural tissue.

- Neuroglia are the supportive cells of neural tissue. They make up more than half the volume of the neural tissue. Neurons, the unit of neural system, are excitable cells.
- Cartilage and blood are specialised type of connective tissues.
- Adipose tissue is a type of loose connective tissue.

161. Answer (3)

Arthritis is inflammation of joints.

Myasthenia gravis is an auto immune disorder affecting neuro-muscular junction leading to fatigue, weakening and paralysis of skeletal muscles.

Muscular dystrophy is the progressive degeneration of skeletal muscles mostly due to genetic disorder.

Tetany is rapid spasms in muscles due to low Ca^{++} in body fluids.

162. Answer (3)

All chordates are divided into three subphyla – Urochordata, Cephalochordata and Vertebrata. In subphylum Vertebrata, notochord is replaced by bony or cartilaginous vertebral column in adults. Therefore, all vertebrates are chordates but all chordates are not vertebrates.

163. Answer (2)

Genetically engineered lymphocytes are not immortal cells and die after some time.

Option (1) is not the answer as the lymphocytes from patient's blood are grown in culture, outside the body but it is not the correct reason.

In option (4), if the gene isolated from bone marrow cells producing ADA is introduced into cells at early embryonic stages, it could be a permanent cure.

164. Answer (3)

Number of base pairs $\times$ distance between 2 consecutive base pairs = Length of DNA molecule
 $N \times 0.34 \times 10^{-9} \text{ m} = 1.1 \text{ m}$

$$N = \frac{1.1}{0.34 \times 10^{-9}} \\ = 3.6 \times 10^9 \\ \approx 3.3 \times 10^9 \text{ bp}$$

165. Answer (4)

Meiosis involves two sequential cycles of nuclear and cell division called meiosis-I and meiosis-II but only single cycle of DNA replication.

The stage between two meiotic divisions is called interkinesis and is generally short lived and involves no DNA replication.

166. Answer (3)

If 8 *Drosophila* in a laboratory population of 80 died during a week, the death rate in the population is $\frac{8}{80} = 0.1$ individuals per *Drosophila* per week.

167. Answer (3)

The correct descending order of taxonomic categories in case of animals is

Kingdom → Phylum → Class → Order → Family → Genus → Species

168. Answer (2)

Birds (*Pavo*) excrete nitrogenous wastes as uric acid in the form of pellet or paste with a minimum loss of water.

Many bony fishes (like *Hippocampus*) and aquatic amphibians *Salamandra* are ammonotelic in nature.

Mammals (like *Ornithorhynchus*) mainly excrete urea and are called ureotelic animals.

169. Answer (3)

Bioprospecting means exploring molecular, genetic and species-level diversity for products of economic importance

170. Answer (4)

Spindle fibres attach to the kinetochores of chromosomes.

Kinetochores are the disc shaped structures present on sides of primary constriction or centromere of chromosomes.

171. Answer (4)

Oogenesis is initiated during the embryonic development stage when a couple of million gamete mother cells (oogonia) are formed within each foetal ovary.

No more oogonia are formed and added after birth in a human female.

At puberty, only 60,000 to 80,000 primary follicles are left in each ovary, rest degenerate during the phase from birth to puberty.

172. Answer (4)

Detritivores (eg. earthworm) break down detritus into smaller particles. This process is called fragmentation.

173. Answer (1)

As the product of 'i' gene binds with the operator region and blocks the transcription and translation of z, y and a genes.

Its product is prevented from binding to the operator by attaching it with the inducer. As the inducer can now no more capable of binding with the repressor, thus, in all the cases, operator always gets attached with the repressor thereby preventing the transcription and transmission of z, y and a.

Even in the presence of lactose, transcription and translation of z, y and a would not occur.

174. Answer (3)

Bioactive molecule, cyclosporin A, that is used as an immunosuppressive agent in organ transplant patients, is produced by the fungus, *Trichoderma polysporum*.

175. Answer (4)

Every 100 mL of oxygenated blood can deliver around 5 mL of O₂ to the tissues under normal physiological conditions.

Every 100 mL of deoxygenated blood delivers approximately 4 mL of CO₂ to the alveoli.

176. Answer (2)

Coagulum or clot is formed mainly of a network of threads called fibrins.

RBCs are destroyed in the spleen. So, spleen is known as the graveyard of erythrocytes.

177. Answer (4)

Cartilage forming the intervertebral disc is present between the adjacent bones of the vertebral column and it is a type of cartilaginous joint.

Areolar tissue is present beneath the skin and it is a type of loose connective tissue.

Smooth muscles are present in the visceral organs.

Presence of intercalated discs is a characteristic feature of cardiac muscles present in the heart.

178. Answer (1)

The intrauterine device (IUD) presently available as the non-medicated IUDs, is Lippes loop.

Copper releasing IUD are CuT, Cu7 and Multiload 375.

Diaphragms, cervical caps and vaults are included under barrier methods of contraception.

179. Answer (2)

- In both spermatogenesis and oogenesis, haploid gametes are formed. So, (a) is true for both.

- The spermatids are transformed into spermatozoa (sperms) by the process called spermiogenesis. Hence, (b) is true for spermatogenesis only.

- Both spermatogenesis and oogenesis, are controlled by LH and FSH secreted by the anterior pituitary. Hence, (d) is true for both.

- Spermatogenesis is a continuous process that begins at puberty. So, (e) is true for spermatogenesis. Oogenesis, on the other hand, begins during embryonic development of the females.

180. Answer (1)

In frogs, the mouth opens into the stomach which in turn continues as the intestine, rectum and finally opens outside by the cloaca.

181. Answer (2)

Frogs exhibit sexual dimorphism. Male frogs can be distinguished by the presence of sound producing vocal sacs and also a copulatory pad on the first digit of the fore limbs which are absent in female frogs.

182. Answer (1)

Statement II is incorrect as the transformation of spermatids into spermatozoa (sperms) is called spermiogenesis. After this, sperm head becomes embedded in the Sertoli cells and are finally released from the seminiferous tubules by the process called spermiation. Hence, Statement I is a correct statement.

Spermatogenesis is the process of formation of sperms from spermatogonia.

183. Answer (1)

In prokaryotes, ER is absent be it RER or SER.

184. Answer (2)

Two additional chambers like crop and gizzard in alimentary canal are present in birds.

Pavo (Peacock), *Psittacula* (Parrot), *Corvus* (Crow) and *Columba* (Pigeon) are birds.

Catla is a bony fish. *Crocodilus*, *Chameleon* and *Bangarus* are reptiles. *Bufo* is an amphibian and *Balaenoptera* is an aquatic mammal.

185. Answer (1)

Autoimmune disorder is a condition in which the body's defense mechanism recognises its own cells as foreign bodies. Sometimes, due to genetic and other unknown reasons, the body attacks self-cells.

Rheumatoid arthritis is an example of autoimmune disorder where body attacks self cells (synovial membrane).

SECTION - B

186. Answer (2)

Impulse transmission across an electrical synapse is always faster than that across a chemical synapse.

→ Chemical synapses use chemicals for transmission which are known as neurotransmitters.

→ The membranes of pre-synaptic and post-synaptic neurons are in close proximity in an electrical synapse.

→ In an electrical synapse, the transmission of the impulse occurs in the form of electrical current from one neuron to the next neuron.

187. Answer (2)

Acquired immunity is a specific type of defence which is not present at the time of birth.

When our body encounters a pathogen for the first time, then the body elicits the primary immune response.

When there is a subsequent encounter with the same pathogen, secondary or anamnestic immune response is elicited.

188. Answer (1)

From 10 parent *E.coli* cells

1st generation

¹⁵N containing 10 cells → 20 mins → 20 cells (all hybrid) ¹⁵N - ¹⁴N

2nd generation

20 cells (hybrid) → 20 mins → 40 cells
 20 cells hybrid ¹⁵N - ¹⁴N
 20 cells with ¹⁴N

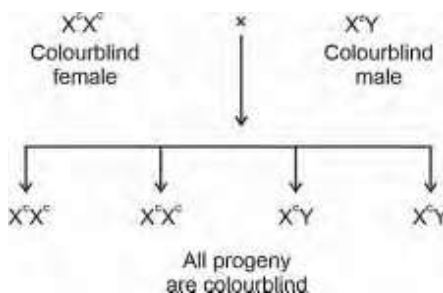
3rd generation

40 cells → 20 min → 20 ¹⁴N containing cells + 20 hybrid and 40 ¹⁴N containing cells
 (20 cells hybrid + 20 ¹⁴N containing cells)

Therefore, after 60 minutes, 60 *E.coli* cells will have DNA totally free from ¹⁵N.

189. Answer (2)

If mother of man is colourblind, then man will also be colourblind as colour blindness is a X-linked recessive trait and shows criss-cross inheritance.



190. Answer (4)

Genetically engineered insulin has A-peptide and B-peptide chains of insulin which are produced separately in *E.coli*, then they are extracted and combined by creating disulphide bonds between them.

Statement (a) is incorrect as genetically engineered insulin does not have an extra stretch of C-peptide.

Statement (c) is incorrect as insulin obtained from cattle and pigs is not genetically engineered.

Statement (d) is incorrect because conversion of pro-insulin to insulin is not required during the production of insulin by genetic engineering as A-peptide and B-peptide chains are produced separately.

Statement (e) is incorrect as allergic reactions to insulin are mostly seen when the insulin is obtained from animals.

191. Answer (4)

Parathyroid hormone is a hypercalcemic hormone *i.e.* it increases the blood calcium levels. It also increases the absorption of calcium from the digested food. Glucocorticoids regulate the carbohydrate metabolism.

Parathyroid hormone stimulates the process of bone resorption.

Reabsorption of Ca^{2+} by renal tubules is a function of PTH.

192. Answer (3)

Ciliated epithelium is mainly present in the inner surface of hollow organs like bronchioles and fallopian tubes. The function of cilia is to move particles or mucus in a specific direction over the epithelium.

Some of the columnar or cuboidal cells get specialised for secretion and are called glandular epithelium. Goblet cells are unicellular glands.

Tendons are dense regular connective tissues. They attach skeletal muscles to bones.

Adipose tissue is a type of loose connective tissue located mainly beneath the skin. The cells of this tissue are specialised to store fats.

193. Answer (4)

- Diaphragms are barrier methods of contraception. They cover the cervix and block the entry of sperms.
- Contraceptive pills are preparations containing either progestogens alone or combination of progestogen and oestrogen. They inhibit ovulation and implantation as well as alter the quality of cervical mucus to prevent entry of sperms.
- Intra uterine devices increase the phagocytosis of sperms within the uterus.

- Lactational amenorrhoea is a natural method of contraception and it is based on the fact that the ovulation and therefore the menstrual cycle do not occur during the period of intense lactation following parturition.

194. Answer (2)

Cloning vectors are the carriers of the desired gene in the host cell. The features desirable in a cloning vector are:-

- Presence of origin of replication
- Presence of marker genes
- Presence of very few, preferably single recognition site for the commonly used restriction enzymes

195. Answer (2)

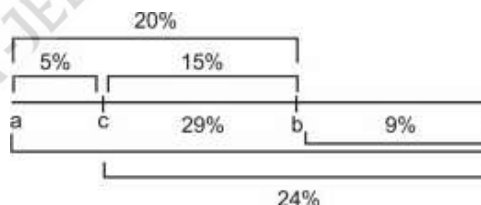
The untranslated regions (UTRs) are present at both 5'-end (before start codon) and at 3'-end at 3'-end (after stop codon).

When oxygen is inadequate for cellular respiration pyruvic acid is reduced to lactic acid by lactate dehydrogenase. The reducing agent is $\text{NADH} + \text{H}^+$ which is reoxidised to NAD^+ .

196. Answer (2)

1% recombination frequency = 1 centi Morgan

To place the genes on a linear chromosome, decreasing order of recombination frequency will be considered.



197. Answer (3)

Slime moulds are classified under kingdom Protista.

Mycoplasma lack cell wall.

Bacteria can be autotrophic as well as heterotrophic.

198. Answer (2)

Glycogen is a polysaccharide and is a storage product in animals.

- Globulins form antibodies which are also known as immunoglobulins.
- Steroids form hormones like testosterone.
- Thrombin is a biocatalyst which converts soluble fibrinogen to insoluble fibrin.

199. Answer (1)

During joint diastole, blood moves freely from atrium to ventricle as atrioventricular valve remains open during joint diastole.

Decrease in ventricular pressure, during ventricular diastole, closes the semilunar valves to produce 'dub' heart sound.

SA node generates the action potential to stimulate atrial contraction.

Bicuspid and tricuspid valves open due to pressure exerted by the blood present in atria and decrease in pressure in ventricles during ventricular diastole.

200. Answer (2)

Flippers of penguins and dolphins are analogous organs as they help in swimming but do not have the same structure.

Homologous organs have the same structure but have different functions according to the needs of the organisms. Hence, homology indicates common ancestry.

Analogous structures have developed for the same function but do not show a similarity in structure. Hence, they are a result of convergent evolution.

Sweet potato is a root modification for food storage whereas potato is an underground stem modification for storage. Hence, they are analogous.

